

Computer Science & IT

Database Management System



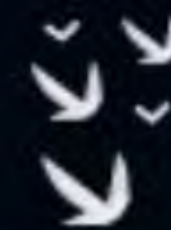
**File organization
And
Indexing**

Lecture No. 03



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Recap of Previous Lecture



✓ **Topic**

Index file

✓ **Topic**

IO cost with indexing

✓ **Topic**

Categories of index

Topics to be Covered



- ✓ **Topic** Types of index
- ✓ **Topic** Primary index
- ✓ **Topic** Clustering index
- ✓ **Topic** Secondary index
- ✓ **Topic** Multi level index





Topic : Types of Index

Single level index

- Primary index ✓
- Clustering index ✓
- Secondary index ✓

Multi-level Index

Dynamic Multi-level index

- B tree
- B+ tree



Topic : Primary Index

If ① Search key attribute is a key {ie. Unique values}
and ② File is ordered based on that search key attribute,
then Index file created on such search key
is called primary index.



Topic : Primary Index

✓ Eid is a key attribute in EMP table. B_1
And
★ Records of the EMP table are ordered based on Eid B_2

• Index is a Primary index.

Search key (Eid)	Pointer
1	
9	
13	
20	
27	
-	

Search key $\rightarrow$ Eid	Unique E.M.P.
1	
7	
9	
10	
13	
16	
20	
24	
27	
30	

Index file (Primary index)

{ It may be Sparse or dense } { In general, Primary index is Sparse }



Topic : Clustering Index

If ① Search key attribute is non-key attribute { i.e. not unique values }
and ② File is ordered on search key attribute,
then index file created on such search key is called
a Clustering index



Topic : Clustering Index

Search Key

not unique
EMP

⊕ Dept-id is a non-key attribute in EMP table

4
Records of the EMP table are physically ordered based on Dept-id

∴ Index is Clustering index

Search Key
Dept-id
Pointer

Dept-id	Pointer
1	
2	
4	
5	
7	

Points to the 1st record corresponding to the search key.

Index file (Clustering)

{ Clustering index will always be Spanned index }

Dept-id	Eid	- - - -
1	7	
2	13	
2	1	
2	5	
4	4	
5	9	
5	15	
7	6	
7	20	
7	12	

B₁
B₂
B₃
B₄
B₅

Block Anchor



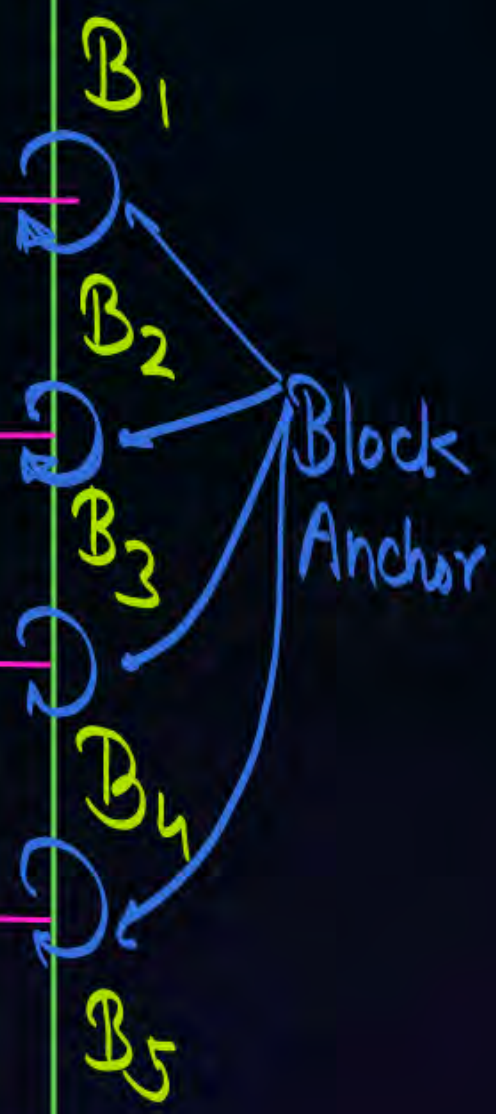
Topic : Clustering Index

EMP

Search key
Dept-id
Pointer

1
2

Dept-id	Eid	- - - -
1	7	
2	13	
2	1	
2	5	
4 ^x	4	
5	9	
5	15	
7	6	
7	20	
7	12	



⊕ Dept-id is a non-key attribute in EMP table

Records of the EMP table are physically ordered based on Dept-id

∴ Index is Clustering index

This is add^r of the first record with search key = 2
If the record is the last record of the block, then we don't know whether it is the last record w.r.t. search key = 2 or not.

So we will use block anchor
And we will access the next block of the file to check if there are more records w.r.t. search key = 2 or not

Note :- $\rightarrow$ ① Primary index can be sparse index or dense index { generally, Primary index is sparse }

② Clustering index is always a Sparse index.

③ At most one primary index is possible on a database file { because file must be ordered based on search _{key} }

④ At most one Clustering index is possible on a database file. { because file must be ordered based on search key }

Note ⑤ On a database file we can either have a Primary index or we can have a Clustering index but not both.
Because at a time file can be ordered w.r.t. At most one attribute



Topic : Secondary Index

- If file is unordered based on search key, then index file created on such search key is called secondary index

Search key values may or may not be unique in database file



Topic : Secondary Index

file is unordered
based on Pass-No.
∴ Secondary index

Pass-No	Pointer
1254	
6102	

file is unordered
based on rating
∴ Secondary index

Rating	Pointer

Pass-No.	Flyer database	Rating
4		1
1		2
9		3
2		2
13		5
7		3
10		5
12		6



Topic : Secondary Index

file is unordered
based on Pass-No.
∴ Secondary index

Pass-No.	Pointer
1254	
61029	

file is unordered
based on rating
∴ Secondary index

Rating	Pointer
1254	
61029	

Pass-No.	Flyer database	Rating
4		1
1		2
9		3
2		2
13		5
7		3
10		5
12		5

Note:- When file is unordered based on search key. then our first objective will be to provide an order using index file so that binary search can be applied.

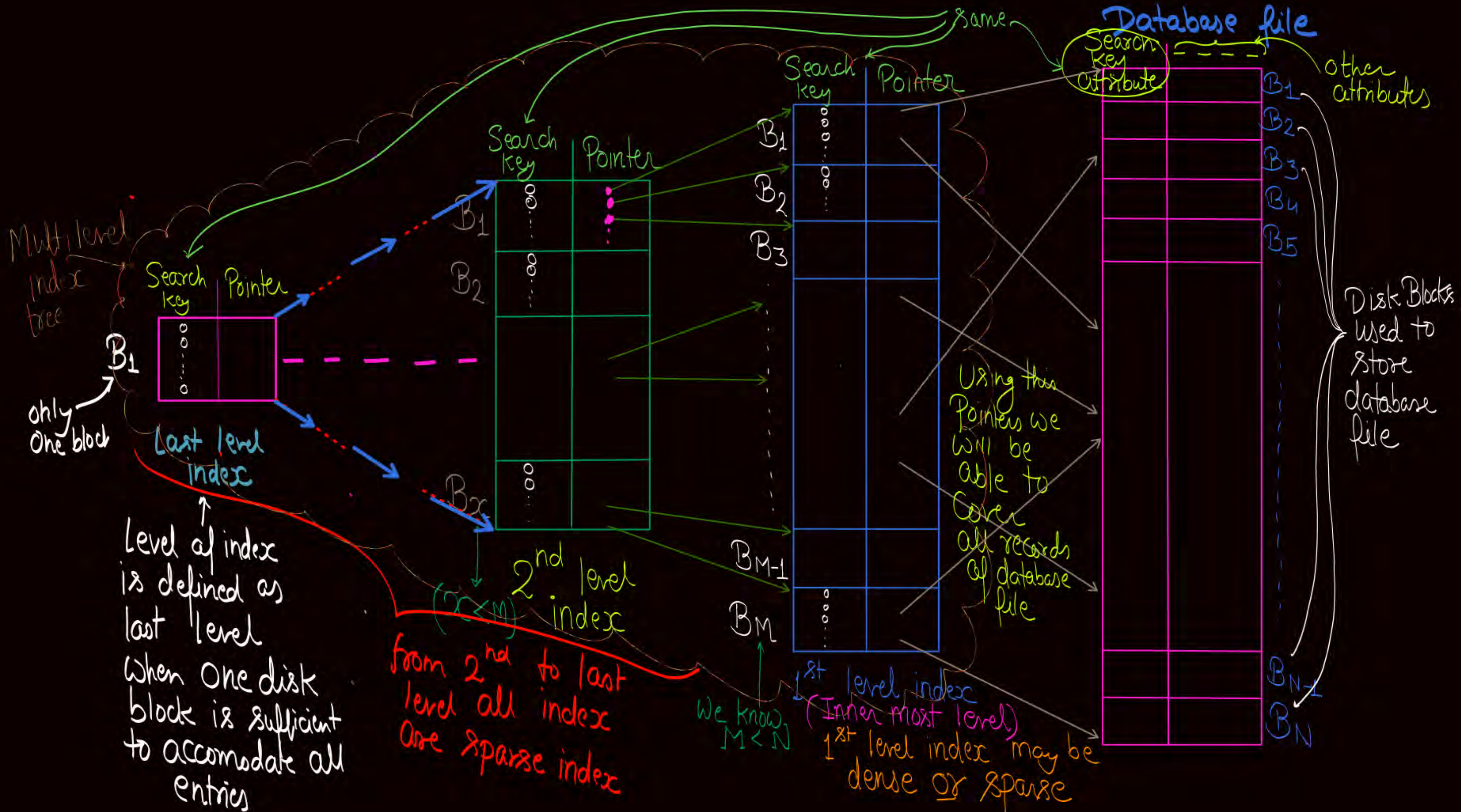
- Secondary index is always a dense index.

Note:- Any number of secondary index are possible on a database file { Because file don't need to be ordered }



Topic : Multi-level Index

- Multilevel index is used to reduce the IO Cost w.r.t single level index



Note:- Multi level index can reduce the IO cost significantly.

$$\text{IO Cost using multi-level Index} = \underbrace{\text{No. of levels in multi level index tree}}_{\substack{\text{from each level of} \\ \text{index tree, exactly} \\ \text{one block will be} \\ \text{access from secondary} \\ \text{memory to main memory}}} + \underbrace{1}_{\substack{\text{1 Block of database} \\ \text{file will be access} \\ \text{from secondary memory} \\ \text{to main memory} \\ \text{in order to access} \\ \text{the record corresponding} \\ \text{to the search key}}}$$



Topic : Multi-level Index

Using multi level index

$$\text{IO Cost} = \text{No. of levels} + 1$$

↳ for all access
{ Best case, worst case, avg case }
all same



Topic : Multi-level Index

In multi-level index,

- ① 1st level of index may be dense or may be sparse
- ② Except first level, all other levels of index are always sparse index



2 mins Summary



✓ **Topic**

Types of index

✓ **Topic**

Primary index

✓ **Topic**

Clustering index

✓ **Topic**

Secondary index

✓ **Topic**

Multi level index



THANK - YOU